Adaptive Experimental Design: a maintained reproduction

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1 What this repository is

A maintained reproduction of Molly Offer-Westort, Alexander Coppock, and Donald P. Green, "Adaptive Experimental Design: Prospects and Applications in Political Science," *American Journal of Political Science* 65(4): 826–844.

The article's replication archive is on the Harvard Dataverse. This repository does not redistribute it. Instead it carries a manifest of the deposit's 24 files with their published checksums, a script that fetches and verifies them, a rewrite of the analysis in current R, a ground truth laying every published number against what the rewrite and the deposited code produce, and this report.

Drafted by Claude Opus 5 under the supervision of Alex Coppock.

Resource	Link
Article	https://doi.org/10.1111/ajps.12597
Replication archive	https://doi.org/10.7910/DVN/CMUHBU
Pre-analysis plan, study 2	https://osf.io/6p7ry
Pre-analysis plan, study 3	https://osf.io/xp8mc

1.1 Layout

```
download_original.R   fetch the deposit and verify it, on every run
run_all.R             the whole pipeline, in order
original/             the deposit (not redistributed; fetched, not committed)
original_manifest.csv file names, sizes and published checksums
maintained/           the rewrite: one script per figure, table or in-text quantity
maintained/output/    everything the rewrite writes, committed so a fresh run can be diffed
maintained/in_text_claims.R  the second instrument: one block per published number
ground_truth/         the extraction, the comparison, and the deposit's own output
README.qmd            this report
errata.qmd            three sentences in the article the data do not support
```

1.2 How to reproduce

Download this repository, open `offer-westort_coppock_green_2021.Rproj`, and run

```r
source("run_all.R")
```

The first run downloads about 132 MB from the Dataverse, almost all of it two saved simulation objects. Later runs verify what is already on disk and download nothing. The whole pipeline took 14 minutes on the machine that produced the committed output, maintained/text_study2_f_tests.R alone accounting for 3.8 of them. Timings are not comparable across machines.

2 The paper

Response-adaptive designs shift assignment probabilities toward whichever arm is performing best, on the argument that a multiarm trial finds its winner faster that way. The article introduces batch-wise Thompson sampling to political science, proposes a control-augmented variant that also protects the control arm so the best arm's treatment effect can be estimated precisely, and evaluates both against a balanced static design.

The evidence comes in three layers. Simulations put nine arms through ten batches of a hundred subjects each under three scenarios: a clear winner, no clear winner, and a competing second best. Study 1 runs Thompson sampling on the wording of minimum wage and right-to-work ballot measures with 1,000 MTurk subjects. Study 2 runs the control-augmented algorithm on six ways of encouraging survey respondents to answer factual questions about the economy accurately, separately among Democrats and Republicans. A third study in the supplement applies a model-based version to a 192-arm conjoint on campaign finance.

3 The deposited archive

3.1 What it contains, and what it writes

The deposit is 24 files: eleven analysis scripts, a library of shared functions, four cleaned data files, two saved simulation objects, four questionnaire codebooks, and a README. It is flat, with no directory structure, and nothing in it was ingested into a derived tabular representation, so `?format=original` and the plain download return the same bytes.

The first thing worth knowing about it is that it writes almost nothing. Of the 46 calls in the deposited code that touch a file, 40 are commented out and 6 are live. The README lists 27 output files; the code produces 6, and three of those four write files the deposit itself ships. Every table and figure in the article therefore exists in the deposit only as console output or as a plot drawn to the active device. That decides how a comparison against it is possible at all, and it is why `ground_truth/extract_archive_values.R` runs the deposited scripts and parses what they print rather than reimplementing their formatting.

3.2 Whether it runs

`ground_truth/run_archive.R` runs every deposited script twice: once against the deposit as downloaded, and once against a copy stripped of the three objects the deposit's own code can regenerate. Both passes run in scratch copies, never in `original/`, which matters here rather than in principle: `simulations_iterated.R` writes `simulations_supplement1.rds` and `simulations_supplement2.rds` and `study_1_rtw-mw_simulations.R` writes `study_1_sims.rds`, and all three are deposited members.

Table 1: Every deposited script, run as shipped and run from a copy stripped to data plus code.

Pass	Script	Completed	Stopped at
As shipped	simulations.R	yes	
As shipped	simulations_iterated.R	no	Error:
As shipped	study_1_rtw-mw_analysis.R	yes	
As shipped	study_1_rtw-mw_F.R	yes	
As shipped	study_1_rtw-mw_simulations.R	yes	
As shipped	study_2_misperceptions_probabilities.R	no	Error in file(file, ...) : argument "file" is missing, with no default
As shipped	study_2_misperceptions_analysis.R	no	Error in readRDS(con, refhook = refhook) : cannot open the connection
As shipped	study_2_misperceptions_additional_analysis.R	no	Error in if (is.na(s)) { : the condition has length > 1
As shipped	study_2_misperceptions_bols.R	yes	
As shipped	study_2_misperceptions_F.R	yes	
As shipped	study_3_conjoint_analysis.R	no	Error in 'repaired_names()':
Stripped	simulations.R	yes	
Stripped	simulations_iterated.R	no	Did not finish within 600 seconds.
Stripped	study_1_rtw-mw_analysis.R	yes	
Stripped	study_1_rtw-mw_F.R	yes	
Stripped	study_1_rtw-mw_simulations.R	no	Did not finish within 600 seconds.
Stripped	study_2_misperceptions_probabilities.R	no	Error in file(file, ...) : argument "file" is missing, with no default
Stripped	study_2_misperceptions_analysis.R	no	Error in readRDS(con, refhook = refhook) : cannot open the connection
Stripped	study_2_misperceptions_additional_analysis.R	no	Error in if (is.na(s)) { : the condition has length > 1
Stripped	study_2_misperceptions_bols.R	yes	
Stripped	study_2_misperceptions_F.R	yes	
Stripped	study_3_conjoint_analysis.R	no	Error in 'repaired_names()':

6 of 11 scripts complete as shipped. Stripping the saved simulations costs two more: `simulations_iterated.R` and `study_1_rtw-mw_simulations.R` both stop trying to regenerate objects that take hours. The four failures common to both passes are worth separating, because they are different kinds of problem.

- `study_2_misperceptions_probabilities.R` calls `write_rds(pmat, path = 'study_2_probabilities.rds')`. The `path` argument was deprecated in readr 1.4.0 and is now an error, so the script dies at its only write.
- `study_2_misperceptions_analysis.R` then fails on the first line that reads that file. The object it needs exists in no deposited file and nothing else creates it, so the analysis behind Figures 5, 6 and F.8 cannot run from the deposit as it stands.
- `study_2_misperceptions_additional_analysis.R` reaches its `stargazer()` call and dies inside it. `stargazer` has not been updated since 2022 and its internals no longer work on current R.
- `study_3_conjoint_analysis.R` fails in `as_tibble_row()` with a name-repair function that returns 192 names for a one-column input, which current tibble rejects.

`simulations_iterated.R` fails in a fifth way that is easy to miss: it builds and prints all four of its tables and only then dies, in the figure code below them, on a `facet_grid(facets = )` argument that was deprecated in ggplot2 2.2.0 and is now defunct. Everything above that point has already printed, which is why the deposit's own Table 1 and appendix table values survive its failure and can be recovered.

3.3 Whether the numbers reproduce

Running is not reproducing, and the two questions have different answers here. Where the deposited code reaches a published number, it returns it. `study_1_rtw-mw_F.R` prints the two randomization inference p-values the study 1 results section reports, 0.429 and 0.199, exactly. `study_1_rtw-mw_analysis.R` prints the eighteen arm counts behind Figures 3 and 4, exactly. `study_2_misperceptions_F.R` prints two p-values that are exactly the two the article reports, but attached to the opposite parties; that finding is below. The four simulation tables print six rows each, and every one of those cells agrees with the article.

4 Errata

Three sentences in the article state a number or a treatment label the data do not support. Each is a misstatement in the prose; every figure and table is correct as printed, and no conclusion changes. They are set out in `errata.qmd`, which renders to `offer-westort_coppock_green_2021_errata.pdf` with its corrected values computed at render time.

1. **The estimated mean of the leading minimum wage arm.** The article gives the winning arm "a probability of being best of 0.219 and an estimated mean of 0.895 over 183 respondents". The arm with a 0.219 probability and 183 respondents is Proposal 3 (N), whose mean is 0.865. The 0.895 belongs to Proposal 3 (Y), a different arm. The same sentence labels the arm "(B" where Figures 3 and 4 use "(N)".
2. **The arm carrying the 4.1-point Republican effect.** The article declares "the 4.1-point average effect of the accuracy treatment statistically significant ($p = 0.014$)". The 4.1-point effect is the Lottery arm's, as the two preceding sentences and Figure 6 both say. The Accuracy arm's Republican estimate is in the opposite direction.
3. **A share rounded twice.** The article reports picking the best minimum wage proposal "37% of the time in static experiments". The simulations give 36.46 per cent, which is 36 at whole-percentage precision; rounding to one decimal first and then again produces 37.

4.1 Two findings that do not belong in an errata

The two study 2 randomization inference p-values are transposed. The article reports p = 0.028 for the Republican trial and p = 0.013 for the Democratic one. The deposited script `study_2_misperceptions_F.R` returns 0.028 for the Democrats and 0.013 for the Republicans. The article's own F statistics settle which way round is right without appealing to the deposit at all: the Republican F is the larger of the two, so the Republican p-value must be the smaller. The values are correct and the labels are swapped. This stays out of the errata because a corrected sentence has to print a number, and the corrected numbers are simulation estimates of 1,000 draws each that the deposit and this reproduction do not agree on to the third decimal: the reproduction returns 0.027 where the deposit returns 0.028, and 0.011 where the deposit returns 0.013. Both readings reject the null at conventional levels, so no conclusion turns on it.

The static-design p-value of 0.152 is not reproducible. The article compares the Republican Lottery effect under the adaptive design against a simulated static one and gives the latter a p-value of 0.152. That figure comes from a bootstrap of 1,000 resamples which the deposit does not seed reproducibly, and the deposited script that would produce it stops earlier, on the missing saved object described above. Over five seeds the p-value runs from 0.076 to 0.117. The estimate and standard error printed beside it on Figure F.8, 0.040 and 0.025, both reproduce.

5 The ground truth

`ground_truth/offer-westort_coppock_green_2021_ground_truth.csv` has 1052 rows. 981 carry a value comparison against the article and 977 of those agree at the precision the page prints. 29 more carry a truth value rather than a number, of which 27 hold. Nothing in the file is typed except the published values, which are read from the article and the supplement; every other column is computed by `ground_truth/build_ground_truth.R`, which `run_all.R` runs.

Table 2: Every row on which the article, the deposit and the reproduction do not agree.

Claim	Article	Reproduction	Deposit	Locus
Minimum wage leading arm's estimated mean	0.895	0.865		paper_internal
Share of simulated static minimum wage trials picking the best proposal, per cent	37	36.460		paper_internal
Republican Lottery p-value, simulated static design	0.152	0.117		archive
The article names the 4.1-point Republican effect as the accuracy treatment's	1	0.000		paper_internal
Republican randomization inference F-test p-value	0.028	0.011	0.013	paper_internal
Democratic randomization inference F-test p-value	0.013	0.027	0.028	paper_internal

`defect_locus` records where the fault lies, because a disagreement reads as a failure of the reproduction and here never is. Four of the six rows are the article disagreeing with its own materials, one is a bootstrap the deposit cannot support, and one is the transposed pair described above.

5.1 Float coverage

Every published float gets rows in the ground truth or a stated reason it cannot. The reason is itself a claim, so "prints no numbers" is recorded only where the published page's text layer confirms it.

Table 3: Published values per float and the share the ground truth reaches.

Float	Published values	Covered	Share	If uncovered, why
TABLE 1	39	39	100%	
TABLE 2	0	0		Study 1 treatment wording. Nothing in the deposit computes it.
TABLE 3	0	0		Study 2 question wording and the official statistics quoted in it. Nothing in the deposit computes them.
TABLE 4	0	0		Study 2 treatment wording. Nothing in the deposit computes it.
FIGURE 1	10	10	100%	
FIGURE 2	0	0		Two views of the same nine simulations as Figure 1; the article states no number from it that Figure 1 does not.
FIGURE 3	0	0		A trajectory plot with no numbers on its face; its endpoints are the posterior probabilities the results section states.
FIGURE 4	36	36	100%	
FIGURE 5	0	0		A trajectory plot with no numbers on its face.
FIGURE 6	40	40	100%	
TABLE C1	24	0	0%	An analytic enumeration in the supplement's worked example. The deposit ships no code for supplement A, B or C.
FIGURE D1	0	0		Plots the cells of Table D.2, which are covered.
FIGURE D2	0	0		Plots the cells of Table D.2, which are covered.
TABLE D2	255	255	100%	
FIGURE D3	0	0		Plots the cells of Table D.3, which are covered.
TABLE D3	165	165	100%	
FIGURE D4	0	0		Plots the cells of Table D.4 and D.5, which are covered, plus the static run at a first batch of 1,000.
FIGURE D5	0	0		Plots the cells of Table D.4 and D.5, which are covered, plus the static run at a first batch of 1,000.
TABLE D4	150	150	100%	
TABLE D5	150	150	100%	
TABLE E6	50	0	0%	Fifty state minimum wage rates transcribed from an external source. The deposited data carry no wage field.
FIGURE F6	20	20	100%	
FIGURE F7	20	20	100%	
FIGURE F8	40	40	100%	
TABLE F7	64	64	100%	
TABLE G8	0	0		Conjoint attribute wording. The level counts it implies are covered separately.
FIGURE G9	0	0		A trajectory plot with no numbers on its face.
FIGURE G10	0	0		A trajectory plot with no numbers on its face; its endpoints are the six probabilities the supplement's results state.

989 of the 1063 numbers the article and supplement print inside floats are covered, across 12 of 14 floats that print any. The two exceptions are Table C.1, an analytic enumeration in a worked example the deposit ships no code for, and Table E.6, fifty state minimum wage rates transcribed from a source the table names.

Five of the covered floats are figures that print an estimate and a standard error beside every point, which makes them published tables in disguise: Figures 4 and 6 in the article, whose published pages have no text layer at all and whose 76 values were read off rendered pages, and Figures F.6, F.7 and F.8 in the supplement.

6 The extraction and the two instruments

`ground_truth/published_claims.csv` is the extraction: 293 numeric claims the article and supplement make, each classified by hand after an exhaustive scan of both documents for numeric tokens and a separate pass for numbers written as words. 251 of them require a claim block.

Table 4: The extraction, by claim type.

Claim type	Needs a block	Claims
definitional	FALSE	33
definitional	TRUE	21
descriptive	TRUE	5
pipeline	TRUE	224
structural	FALSE	4
structural	TRUE	1
transcribed	FALSE	5

The claims that need no block are those no part of the pipeline can reach: payments, field dates, treatment wording, the analytic worked examples in supplement C, and the two conjoint sample sizes, since the deposit ships study 3 as batch-level fitted data rather than a respondent file. Each carries its reason in the file.

Two instruments read the same pipeline output by separate paths. `ground_truth/build_ground_truth.R` assembles the comparison. `maintained/in_text_claims.R` prints one line per claim beside the sentence it belongs to, with the article's own words in a block comment above the code. Neither reads the other's result, and where they disagree one of them is wrong.

The coverage gate at the end of the build runs `in_text_claims.R` as a program in its own environment, captures what it prints, and asserts four things: that the number of claims printed equals the number the extraction requires, that the two sets of identifiers are equal in both directions, that every printed value matches the build's independently derived one, and that the published string re-rendered at each claim's recorded precision is the string the extraction stores. Both files look that precision up in the extraction, so neither can name a different one, which is what makes the value comparison a check rather than an identity.

7 The maintained rewrite

`maintained/` is 32 scripts: a shared `helpers.R`, one per published figure and table, and one per group of in-text quantities. The deposit's `utils.R` is reproduced in `helpers.R` rather than sourced, because it loads its own packages and resolves paths relative to the working directory, and every random-number call is kept in the deposit's order so seeded results stay comparable.

The substitutions the rewrite makes:

Deposited	Maintained	Why
`%>%`	`|>`	Native pipe
`gather()`, `reshape2::melt()`	`pivot_longer()`, an explicit long-form helper	Superseded
`geom_errorbarh(height = 0)`	`geom_linerange()`	Deprecated in ggplot2 4.0.0
`position_dodgev()`	`position_dodge()`	Current ggplot2 dodges along a discrete axis without help
`write_rds(path = )`	`write_rds(file = )`	`path` is an error under readr 2.2.0
`stargazer()`	`modelsummary()`	`stargazer` fails outright on current R
`facet_grid(facets = )`	`facet_grid(rows ~ cols)`	Defunct since ggplot2 2.2.0
`as_tibble_row(.name_repair = )`	assignment into the row	The repair function no longer matches a one-column input

Two of these are not cosmetic. `study_2_misperceptions_probabilities.R` cannot run at all under current readr, so `maintained/study2_probabilities.R` is what makes the study 2 analysis reachable; it takes about 2 minutes at 10,000 Bayesian bootstrap draws. And `study_3_conjoint_analysis.R` cannot run at all under current tibble, so `maintained/study3_probabilities.R` is what makes Figures G.9 and G.10 reachable.

7.1 Porting the deposit's profile grid

Study 3's 192 conjoint profiles are built in the deposit with `expand.grid()`, and the model matrix that indexes them decides which posterior probability belongs to which profile. `expand.grid()` varies its first argument fastest and `tidyr::expand_grid()` varies its last, so a naive port transposes the grid: the numbers come out identical and land on the wrong profiles, which no comparison against the estimates can see. `maintained/study3_probabilities.R` supplies the four attributes in reverse and puts the columns back, and the resulting profile order and model matrix are identical to the deposit's, row for row.

What confirms it is the supplement's own results. The top three adaptive profiles reproduce at 0.338, 0.213 and 0.076, and the top three static profiles at 0.194, 0.161 and 0.140, each carrying the attribute levels the supplement describes. A transposition would have moved every one of them.

7.2 A labelling error in the deposit that changes nothing

`study_2_misperceptions_F.R` renames the treatment levels to `lottery, accuracy, control, google, direction, extratime` and then asks the control-augmented algorithm to protect position 3. The levels in the data are `control, lottery, accuracy, google, direction, extratime`, so position 3 is the accuracy arm rather than the control. Under the sharp null the test is built against, every arm is exchangeable, so which index the algorithm protects changes the random stream and nothing else. `maintained/text_study2_f_tests.R` computes both and writes both.

8 Figure verification

Every figure script writes a CSV of the estimates it plots, so a fresh run can be diffed against the committed one without comparing images. Each rendered figure was also laid beside the published page, which is the only check that catches a plot drawing the right numbers in the wrong places.

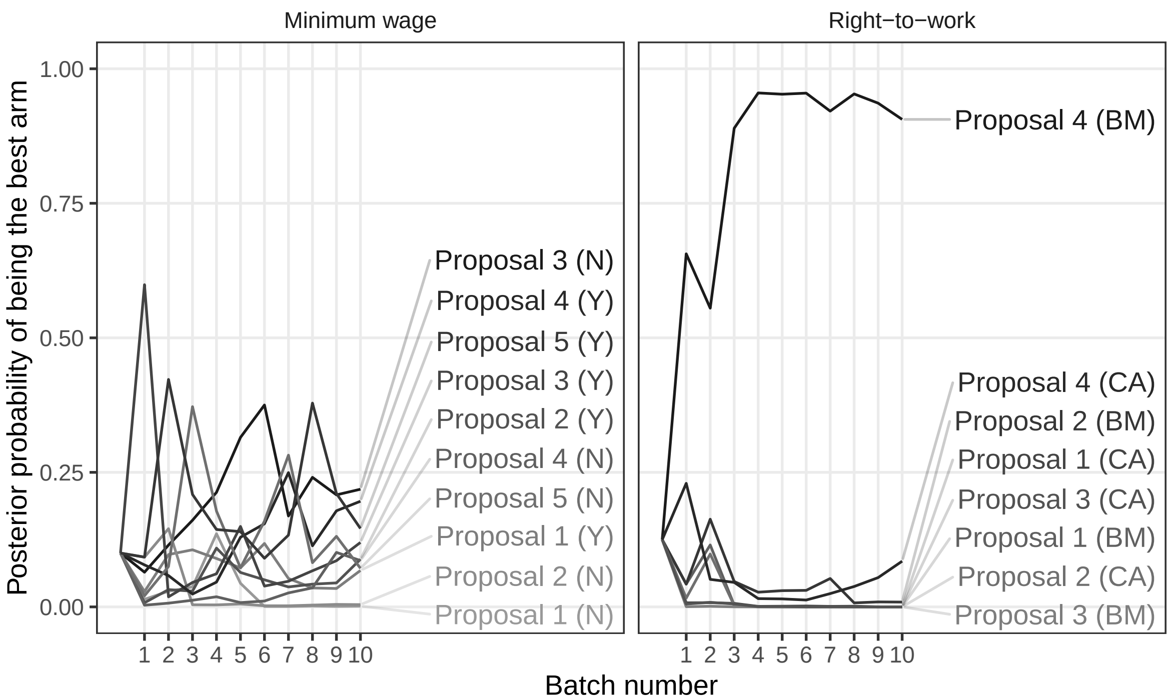

Figure 1: Study 1, over time posterior probabilities (Figure 3).

Two differences from the published figures are deliberate. Appendix Figures D.4 and D.5 carry an eleventh point at a first batch of 1,000, where the whole sample sits in the first batch and there is nothing left to adapt on; the deposit binds its static run in at that position under both algorithm labels, and the rewrite does the same. Figure 6's covariate-adjusted series is drawn with a distinct shape as well as a distinct shade, which the published version does not do.

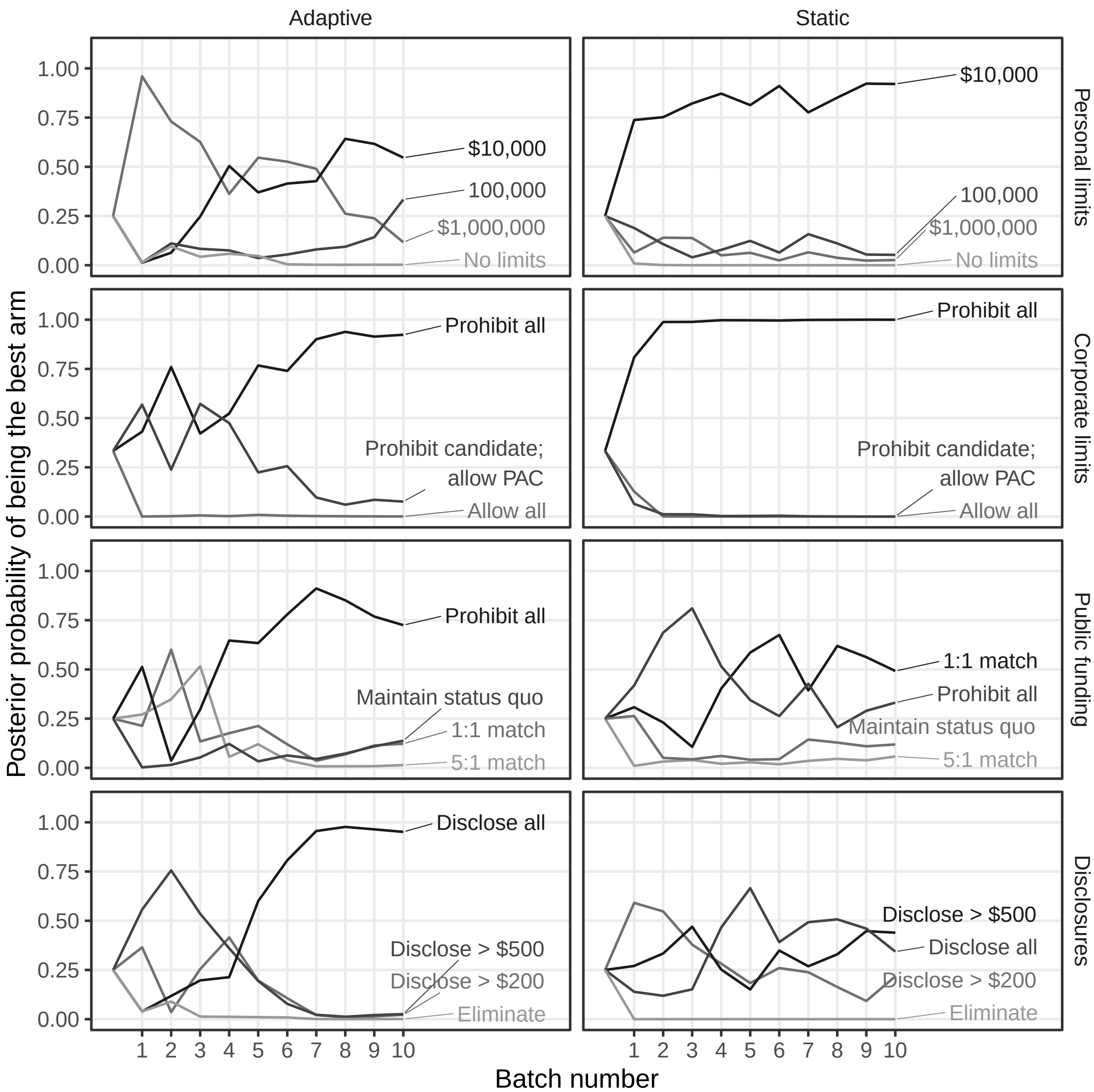

Figure 2: Study 3, over time posterior probabilities by attribute (Figure G.9). This is the figure a transposed profile grid would have scrambled.

9 Rewrite verification

`run_all.R` runs the pipeline end to end, ending with a second verification of `original/`. The deposited archive is checked on every run, before anything else and again afterwards, against the published MD5 and the recorded byte size of all 24 files, and `original/` is asserted to hold nothing the manifest does not list. All 24 published checksums agree with the bytes the Dataverse serves, so the two checksums the archive records are one claim rather than two.

Table 5: The eight slowest steps of one full run.

Script	Minutes
maintained/text_study2_f_tests.R	3.8
ground_truth/extract_archive_values.R	3.2
maintained/study3_probabilities.R	2.2
maintained/study2_probabilities.R	2.1
maintained/figure_f8_simulated_comparisons.R	1.2
maintained/text_study1_f_tests.R	1.1
maintained/text_study2_batched_ols.R	0.2
maintained/simulations_illustrative.R	0.0

`maintained/output/` is committed, so a reader can run the pipeline and diff the result. Everything except the figure PDFs comes back byte-identical; a PDF records the time it was written, so its bytes change on every run and its checksum is not a stable fact about the analysis.

10 R environment

Table 6: The environment the committed output was produced in.

Component	Version
R	4.6.0
tidyverse	2.0.0
estimatr	1.0.6
ggplot2	4.0.3
MCMCpack	1.7.1
bandit	0.5.1
modelsummary	2.6.0

The analysis is seeded throughout at the deposit's own seed, 95126. Three quantities are nonetheless not bit-reproducible against the deposit and are reported as such above rather than presented as agreement.

11 Licence

CC0 1.0, matching the deposit.